K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY

A Project Report on

INTERGALACTIC SPACE TRAVEL REQUEST SYSTEM (ISTR) IN SERVICENOW

Submitted by

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INTERGALACTIC SPACE TRAVEL REQUEST SYSTEM (ISTR)

Complete Technical Specification & Implementation Guide

Application Scope: **Platform**: ServiceNow

x_istr_space_travel

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1. EXECUTIVE SUMMARY & OVERVIEW

The Intergalactic Space Travel Request System (ISTR) is an enterprise-grade scoped application built on the ServiceNow platform. Designed to digitize and modernize the lifecycle of off-world transport requests, ISTR replaces fragmented, manual scheduling processes with a centralized, secure, and fully auditable digital workflow.

The application provides a structured Service Catalog interface for end-users to submit travel requests, while leveraging ServiceNow Flow Designer to orchestrate multi-tiered, role-based approvals, automated risk evaluation, financial verification, and dynamic email notifications. Beyond automating request intake, ISTR establishes a standardized operational baseline for enterprise off-world logistics. By replacing ad-hoc communications with structured catalog variables and system-driven workflow transitions, the platform eliminates procedural bottlenecks, reduces request processing latency, and prevents unauthorized travel assignments. The centralized architecture guarantees that every travel record remains linked to its original submission variables, manager endorsements, and financial authorization history.

Furthermore, ISTR serves as a scalable foundation designed to support future operational expansions, such as automated orbital trajectory scheduling, real-time resource allocation, and multi-system integration. By embedding strict governance, audit logging, and automated notification loops directly into the platform workflow, the application delivers complete end-to-end operational transparency for mission commanders, compliance officers, and finance directors alike.

2. ENVIRONMENT SETUP & TOOLING

The Intergalactic Space Travel Request System (ISTR) was developed by provisioning a ServiceNow Personal Developer Instance (PDI) on the current platform release with full administrative access. Within ServiceNow Studio, a dedicated application scope (x_istr_space_travel) was created to isolate and protect all application artifacts, including custom tables, flow logic, and UI elements. Dedicated security roles (x_istr.user, x_istr.approver, x_istr.finance, x_istr.admin) and user groups (ISTR Requesters, ISTR Approvers, ISTR Finance, ISTR Mission Control) were configured to enforce role-based access control (RBAC). Finally, system email properties and Flow Designer engines were enabled to support end-to-end multi-tier approval routing, impersonation testing, and notification delivery.

To maintain development integrity across the application lifecycle, all metadata configurations, table attributes, security policies, and workflow logic were systematically grouped within a dedicated scoped Update Set. This granular isolation ensures that application artifacts can be captured, validated, and migrated across sub-production and production environments without dependency conflicts or unintended global scope modifications.

Parameter / Layer	Configuration / Value
Platform / Hosting	ServiceNow Cloud Platform (Personal Developer Instance - PDI)
ServiceNow Release	Washington DC / Xanadu Release
Application Scope	Custom Scoped Application (x_istr_space_travel)
Development Tooling	ServiceNow Studio, Flow Designer, Service Catalog Designer
Target Data Model	Custom Table (x_istr_space_travel_request) & Extended Service Catalog Tables (sc_req_item, sys_approver)
Process Engine	Flow Designer (ISTR Approval Flow)
Email Subsystem	ServiceNow Outbound Email Engine & Email Logs (sys_email)

Complementing the platform setup, localized system properties were configured to simulate realistic operational scenarios during testing. Specific parameters—including automated approval engine rules, mail outbound processing, and user impersonation controls—were tuned to mirror production-grade security standards. This tooling setup provided a controlled sandbox environment capable of executing complex end-to-end integration tests without compromising baseline platform integrity.

3. PROJECT OBJECTIVES

The primary objective of the ISTR project is to deliver a robust, automated digital architecture for managing space transport operations. The project achieves this through five key functional targets:

- Standardize Data Capture & Request Entry: Implement a centralized Service Catalog Item (Submit Space Travel Request) equipped with client-side validations, field tooltips, and regex pattern matching to ensure complete submission data.
- Automate Dynamic Multi-Tier Approvals: Engineer a Flow Designer process engine that routes approvals sequentially to Line Managers, Finance Officers, and conditionally to Mission Control based on mission risk parameters.
- Enforce Governance & Role-Based Security: Restrict data access and administrative execution via custom security roles mapped to specialized functional user groups
- Enhance System Navigability: Develop tailored Application Menus(Intergalactic Space Travel) and direct modules for seamless user navigation.
- Provide Visibility & Auditability: Automate dynamic notification emails and embed state tracking across record lifecycles to maintain full operational audit logs.

SYSTEM ARCHITECTURE DIAGRAM

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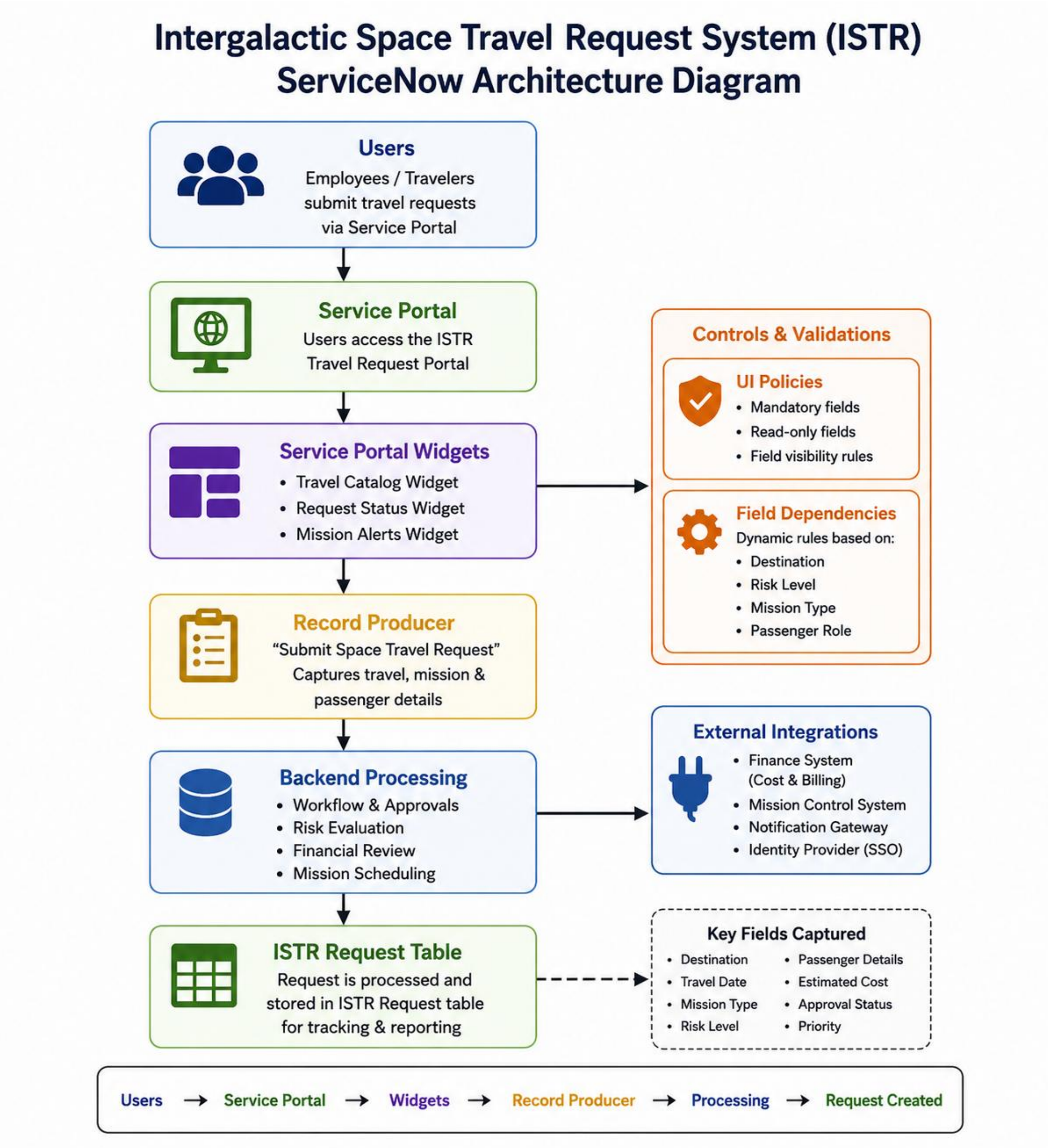

Figure 1.1: Architecture of the Intergalactic Space Travel Request (ISTR)

4. DATA MODEL & ROLE-BASED SECURITY

Application security is enforced through custom security roles mapped directly to platform Access Control Lists (ACLs) and specialized functional groups.

Role Name	**Assigned Group**	**Permissions & Scope**
x_istr.user	ISTR Requesters	Can submit catalog requests and read own created records.
x_istr.approver	ISTR Approvers	Can view assigned approval tasks and perform Line Manager sign-offs.
x_istr.finance	ISTR Finance	Can review/edit financial fields (estimated_cost) and approve Stage 2 tasks.
x_istr.admin	ISTR Mission Control	Full CRUD access, mandatory high-risk approvals, and UI Action execution.

5. FLOW DESIGNER IMPLEMENTATION & VERIFICATION

The workflow engine (ISTR Approval Flow) orchestrates the end-to-end process lifecycle upon submission of the catalog item. During quality assurance testing, critical variable mapping and dot-walking fixes were successfully implemented and validated.

Flow Execution Pipeline (7-Step Architecture):

Step	**Action Name**	**Configuration & Data Mapping Details**	**Status**
1	Get Catalog Variables	Extracts catalog item variables (requester, mission_name, destination, risk_level, estimated_cost).	Completed

2	Look Up User Record	Queries sys_user using dot-walked path: 1 - Get Catalog Variables ➔ requester ➔ Sys ID.	Completed

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3	Ask For Approval (Stage 1 & 2)	Routes sequential approval tasks to Requester's Manager and ISTR Finance group.	Completed
4	If High or Critical Risk	Flow logic decision branch checking if risk_level matches High or Critical.	Evaluated
5	Ask For Approval (Stage 3)	Conditional approval step routed to ISTR Mission Control group for elevated risk missions.	Completed
6	Update Record	Updates target sc_req_item record state to Ready for Launch (or Approved).	Completed
7	Send Email	Dispatches dynamic HTML approval notification email to 1 - Get Catalog Variables ➔ requester ➔ Email.	Completed

8. CONCLUSION

The Intergalactic Space Travel Request System (ISTR) successfully demonstrates a complete, production-grade enterprise application built on the ServiceNow platform. By digitizing off-world transport operations within a dedicated scoped application (x_istr_space_travel), the project replaces manual scheduling with an automated, role-based, and fully auditable digital architecture. Through the strategic integration of Service Catalog variable capture, Flow Designer approval orchestration, and automated state transitions, ISTR achieves seamless end-to-end operational governance. The enforces strict security via four custom roles, routes multi-tier approvals based on real-time risk parameters, and maintains full transparency through dynamic email notifications and complete system execution logs. Ultimately, this app establishes a scalable, error-free foundation for enterprise space logistics that guarantees compliance, accelerates request fulfillment, and delivers full lifecycle.